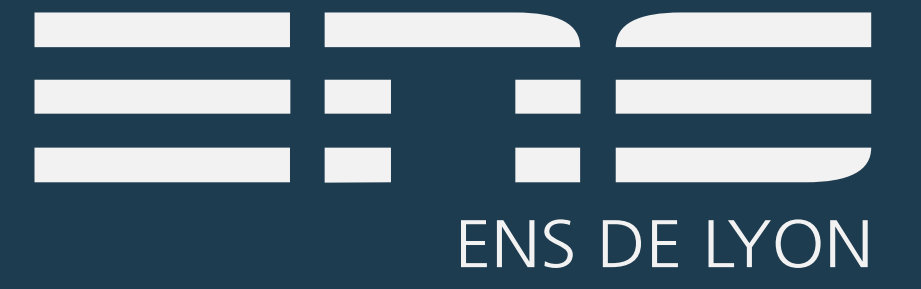


Learning to reconstruct from clipped signals alone

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Summary / TLDR

Goal: Learn to reconstruct clipped signals from measurement data alone.

Challenges:

1. When is learning possible?
2. How can we train a network with clipped measurements only?

Unsupervised learning for declipping problem

Problem setup:

$$\mathbf{y} = \eta(\mathbf{x})$$

where:

- $\mathbf{x}$ belongs to a set $\mathcal{X} \subset \mathbb{R}^n$.
- $\eta(\cdot)$ an element-wise **non-linear** operator defined as:

$$\eta(x_j) = \begin{cases} -\mu & \text{if } x_j \leq -\mu \\ x_j & \text{if } x_j \in [-\mu; \mu] \\ \mu & \text{if } x_j \geq \mu \end{cases}$$

Unsupervised learning:

- **Dataset:** $(\mathbf{x}_i, \mathbf{y}_i)_{i \in I}$ only $(\mathbf{y}_i)_{i \in I}$.
- **Model identification:** Recover the signal set $\mathcal{X}$ given $\mathcal{Y} = \eta(\mathcal{X})$?
- **Signal recovery:** Recover the signal $\mathbf{x}$ from $\mathbf{y}$ with know $\mathcal{X}$?

Amplitude invariance prior: $g\mathbf{x} \in \mathcal{X}$ for all $g \in \mathbb{R}_+^*$ and $\mathbf{x} \in \mathcal{X}$.

- Natural signals are **invariant** to changes in amplitude.
- **Amplitude** invariance enables learning from clipped data.

Exploiting scale invariance

Amplitude invariance:

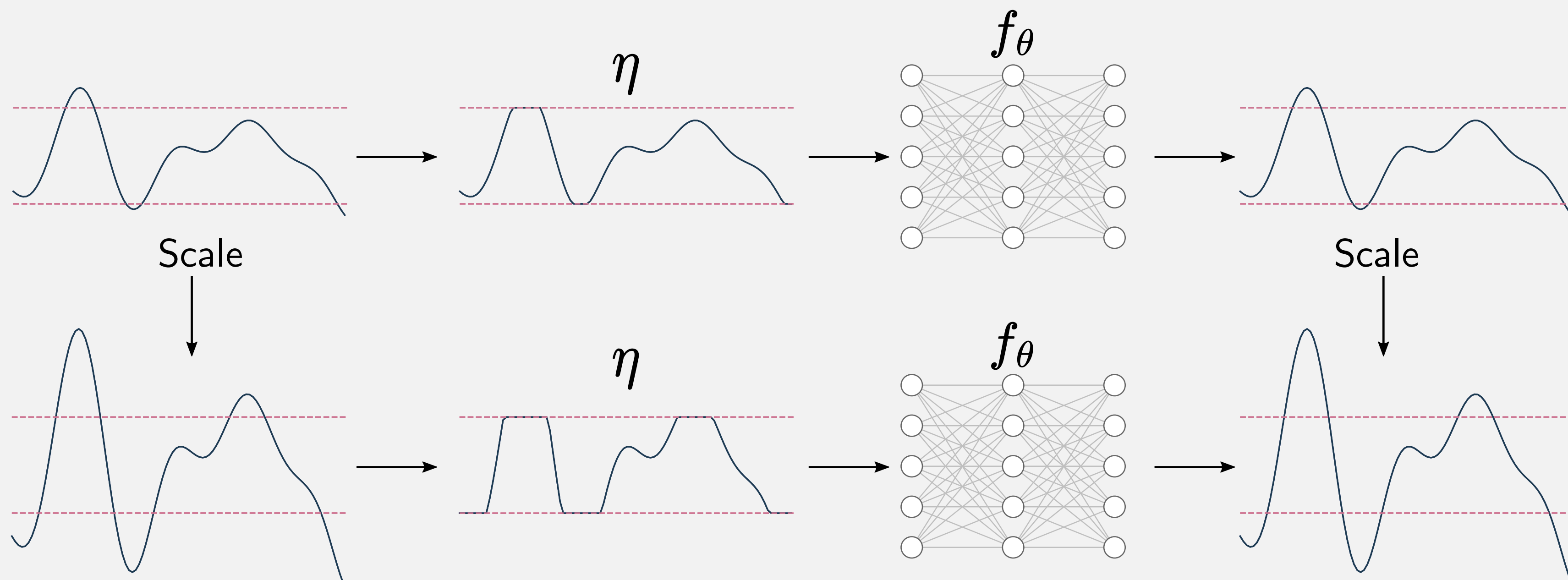
$$\forall \mathbf{x} \in \mathcal{X}, \forall g \in \mathbb{R}_+, g\mathbf{x} \in \mathcal{X}. \quad (1)$$

Key idea:

each measurement can be seen as a signal through another threshold-type operator:

$$\eta(\mathbf{x}) = \eta(gg^{-1}\mathbf{x}) := \eta_g(\mathbf{x}'). \quad (2)$$

with $\mathbf{x}' = g^{-1}\mathbf{x} \in \mathcal{X}$.



Exploiting invariance: $g\mathbf{x} \in \mathcal{X} \implies f_\theta$ has to be able to reconstruct $\mathbf{x}$ and $g\mathbf{x}$ for all $\mathbf{x}$ in $\mathcal{X}$:

$$f_\theta(\eta(g\mathbf{x})) = g\mathbf{x} \text{ and } f_\theta(\eta(\mathbf{x})) = \mathbf{x} \quad (3)$$

Equivariance property:

$f \circ \eta$ equivariant for the multiplicative group $(\mathbb{R}_+, \cdot)$:

$$gf_\theta(\eta(\mathbf{x})) = f_\theta(\eta(g\mathbf{x}))$$

Proposed method

Training loss:

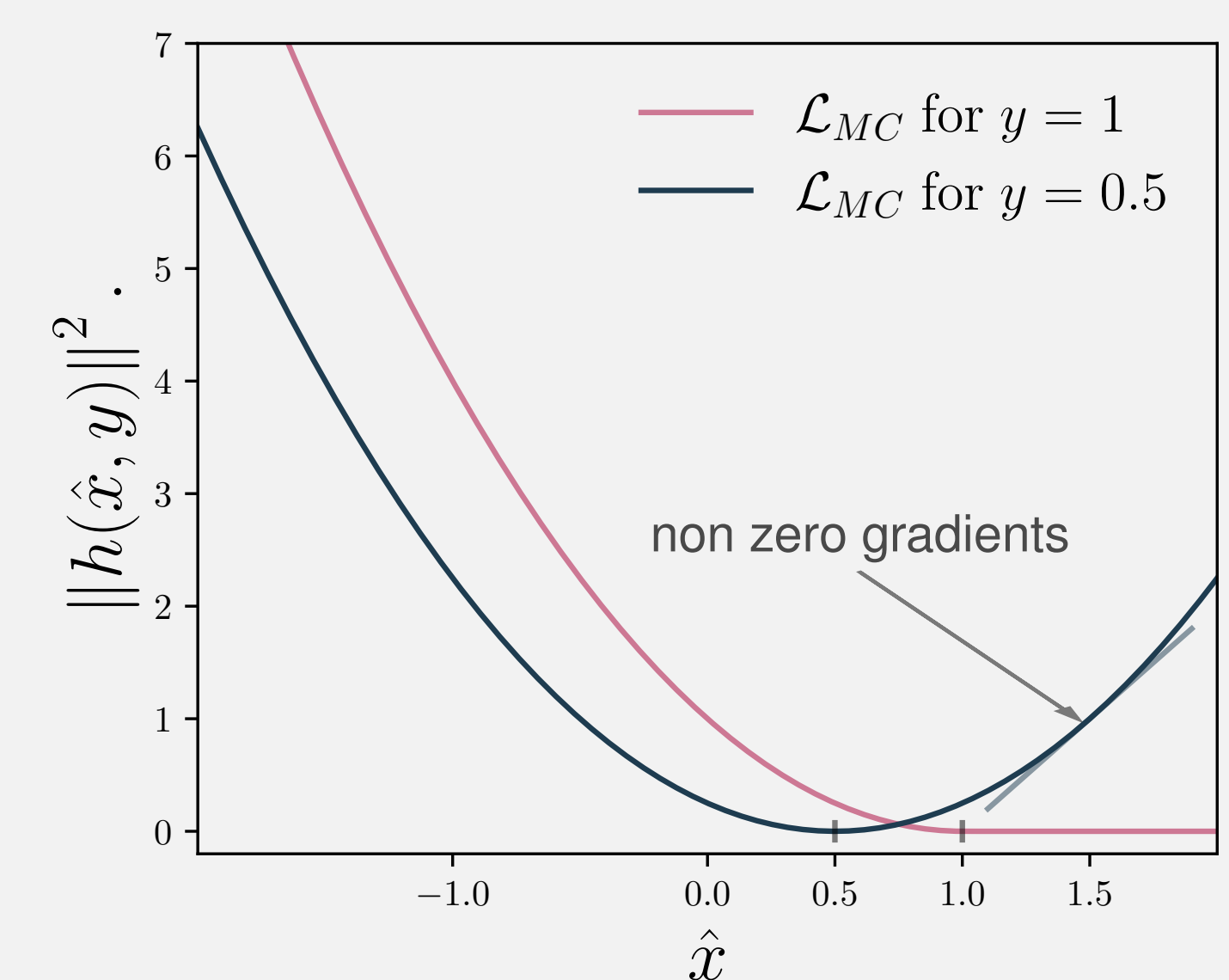
$$\mathcal{L}(\theta) = \mathcal{L}_{\text{MC}}(\theta) + \mathcal{L}_{\text{EI}}(\theta) \quad (4)$$

Measurement consistency loss:

$$\mathcal{L}_{\text{MC}}(\theta) = \sum_{i=1}^N \|h(f_\theta(\mathbf{y}_i), \mathbf{y}_i)\|^2. \quad (5)$$

with h defined as

$$h(\mathbf{u}, \mathbf{v})_j = \begin{cases} v_j - u_j & \text{if } |v_j| < \mu \text{ or } \text{sign}(v_j)u_j < \mu \\ 0 & \text{if } |v_j| = \mu \text{ and } \text{sign}(v_j)u_j \geq \mu. \end{cases}$$



Equivariance promoting loss:

$$\mathcal{L}_{\text{EI}}(\theta) = \sum_{i=1}^N \mathbb{E}_{g \sim p_g} [\|gf_\theta(\mathbf{y}_i) - f_\theta(\eta(gf_\theta(\mathbf{y}_i)))\|^2] \quad (6)$$

Synthetic dataset

Synthetic dataset construction:

- Generate a random universal subspace of $\mathbb{R}^{100}$ with dimension d .
- Sample vectors from the subspace.
- Rescale vectors by a factor so that a proportion $v \in (0, 1)$ of measurements are clipped.

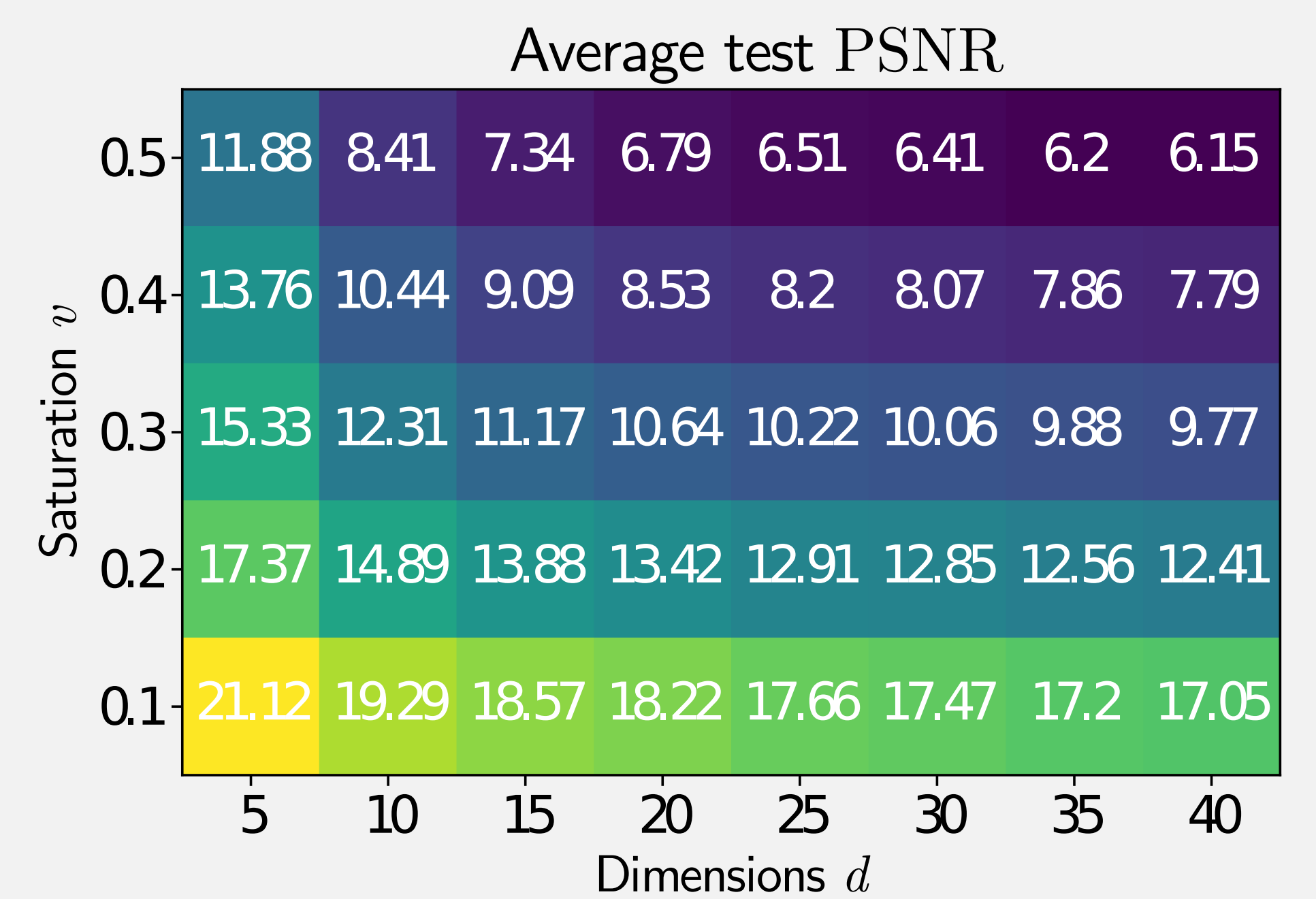


Figure 1: Reconstruction performance as a function of clipped part v and model dimension d .

Real dataset

Musics dataset: 21898 sounds of 1 sec each, 22050Hz.

Methods	PSNR
Identity	5.28 ± 2.23
Social Sparsity	10.25 ± 4.64
Supervised	12.00 ± 2.31
Unsupervised	11.41 ± 3.02

Robustness to distribution shift:

- **Supervised** train on music dataset (with ground-truth).
- **Unsupervised** train on music + voice dataset (without ground truth).

Methods	PSNR
Identity	6.54 ± 2.34
Supervised	10.94 ± 2.00
Unsupervised	11.92 ± 2.46

Implementation details

- **Bias-free network:**

$$f_\theta(\alpha\mathbf{y}) = \alpha f_\theta(\mathbf{y}) \quad \forall \alpha \in \mathbb{R}_+. \quad (7)$$

- **Masking:**

$$\hat{x}_j = (1 - b_j)y_j + b_j f_\theta(\mathbf{y})_j. \quad (8)$$

$$\text{with } b_j = \frac{\max(0, |y_j| - \tau\mu)}{1 - \tau\mu}.$$

References

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- [2] Sreyas Mohan, Zahra Kadkhodaie, Eero P Simoncelli, and Carlos Fernandez-Granda. Robust and interpretable blind image denoising via bias-free convolutional neural networks. *arXiv preprint arXiv:1906.05478*, 2019.
- [3] Dongdong Chen, Julián Tachella, and Mike E Davies. Equivariant imaging: Learning beyond the range space. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021.